Smaller consistency equations through degree separation, congruence norms, and direct proof certificates

Research contribution for the ProveIt repository

23 October 2026

Abstract

We compare the six newly supplied constructions and combine positive-gap source coordinates, degree-separated coefficient packing, calibrated carry deletion and a congruence-separated two-root norm. The resulting fixed integer polynomials improve the operation/degree tradeoffs for nine positive and ten unrestricted integer unknowns. The full literal circuits and coefficient payloads accompany the paper. A distinct proof-certificate source avoids the K215 machine and Jones's universal expression; its local polynomial relation is separated carefully from the still-uninstantiated global trace compiler. All consistency equivalences claimed for completed equations are ordinary mathematical proofs formalizable in ZFC, with identified inherited compiler and number-theoretic dependencies. We claim neither global minimality nor a new end-to-end proof-assistant certificate.

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1 Results and interpretation

The requested target is a fixed ordinary integer polynomial P, on a stated domain D, for which

$$\mathrm{ZFC} \vdash \left(\neg\exists \mathbf{x} \in D^n \ P(\mathbf{x}) = 0\right) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}).$$

Here consistency is syntactic: there is no finite derivation of a contradiction in the fixed effective proof calculus for ZFC. We prove an existence equivalence to such a refutation, then negate it. No argument assumes that ZFC is consistent, and no part of the work decides that question.

Domain	New construction	Operations	Exact degree
$\mathbb{Z}^9_{>0}$	Gap $N - q$, simple majorant	444	1,845,940
$\mathbb{Z}^9_{>0}$	Gap N, simple majorant	449	1,834,516
$\mathbb{Z}^9_{>0}$	Reflected gap, simple majorant	451	1,825,540
$\mathbb{Z}^9_{>0}$	Gap $N - q$, refined majorant	476	1,520,178
$\mathbb{Z}^9_{>0}$	Gap N, refined majorant	481	1,510,770
$\mathbb{Z}^9_{>0}$	Reflected gap, refined majorant	483	1,503,378
$\mathbb{Z}^{10}$	Reflected gap, congruence norm	494	3,507,876
$\mathbb{Z}^{10}$	Gap $N - q$, congruence norm	490	3,547,076
$\mathbb{Z}^{15}_{>0}$	Integral-root transport	88	24
$\mathbb{Z}^{15}_{>0}$	Integral-root transport	87	26
$\mathbb{Z}^{15}_{>0}$	Integral-root identity	78	30

The nine-variable family additionally has explicit nonnegative-domain versions, each with seven extra operations and the same degree. Against the newest incoming nine-positive circuit (473 operations, degree 1,910,404), the 444-operation variant improves both objectives; the refined variants trade some gates for lower degree. Against the incoming ten-integer degree/operation records considered here, both completed ten-integer variants improve substantially. The ten-integer evaluator materializes a fixed threshold constant: retaining the former five constant-only gates would give 499 and 495 operations respectively. This convention does not change either polynomial or its degree.

Every named intermediate register in the completed equations is a polynomial abbreviation, not another quantified variable. Every exponent is a fixed nonnegative integer. There are no divisions, variable powers, square roots, bit operations or unbounded quantifiers inside the final literal circuits. Divisions, radicals and population counts occur only in proofs. The four compiler ports are bound to complete, coherent numerical tuples. Derived large coefficients are materialized as signed binary payloads with hashes rather than left as unspecified universal-polynomial parameters.

What simplicity means here. Number of unknowns, their domain, total polynomial degree, arithmetic circuit cost, and coefficient storage are different objectives. We count a live binary addition, subtraction or multiplication as one operation; multiplication by a numeral is charged, and fixed powers are compiled to multiplication gates. Reusing a previously computed node is free. Fixed integer coefficient construction is not included in this circuit cost. Thus a smaller operation count is not a bit-complexity or running-time claim. We retain the relevant tradeoffs rather than declare a single globally simplest polynomial.

A route beyond the previous architecture. The new direct proof source does not use the K215 transition table or the Jones universal expression. It addresses the user's broader permission to choose another equiconsistent formalism. Its local transition polynomial and accepting-path equivalence

are explicit. The report does not misidentify that local polynomial with a completed consistency equation: converting all finite paths to one fixed ordinary polynomial still has a separately stated implementation boundary.

2 Recent commits and the incoming frontier

The review is pinned to ProveIt main `beb0192cb8dc36f78c369c080f2adfbc1887a486`, with the six original archive bytes pinned at arrival `7331974b696693a8eed2aa5f0f64f1c19d9569f0`. They are placed as contributions 54–59 under https://github.com/VladimirReshetnikov/ProveIt/tree/beb0192cb8dc36f78c369c080f2adfbc1887a486/Computability/DiophantineConsistency/con-zfc-diophantine-forms. The original ZIP identities, retained member hashes and direct-file provenance are in the artifact manifests.

Incoming contribution	Domain	Operations	Degree
54, calibrated carry	$\mathbb{Z}_{>0}^{9}$	564	4,400,764
54, calibrated carry	$\mathbb{Z}^{10}$	554	14,699,152
55, congruence norms	$\mathbb{Z}^{10}$	586	9,152,284
56, free coordinate	$\mathbb{Z}_{>0}^{9}$	560	4,686,812
57, separated degrees	$\mathbb{Z}_{>0}^{9}$	499	2,628,944
57, separated degrees	$\mathbb{Z}^{10}$	544	8,317,400
57, two norms	$\mathbb{Z}_{>0}^{12}$	89	258
57, balanced quotient	$\mathbb{Z}_{>0}^{12}$	94	242
58, integral root	$\mathbb{Z}_{>0}^{15}$	79	30
58, integral root	$\mathbb{Z}_{>0}^{15}$	88	26
58, integral root	$\mathbb{Z}_{>0}^{15}$	90	24
59, degree separated	$\mathbb{Z}_{>0}^{9}$	473	1,910,404
59, degree separated	$\mathbb{Z}^{10}$	533	8,469,528

The table reports the incoming exact-degree claims, except where the independent source audit supplies a fresh degree proof for the 473-operation nine-positive equation. Intake independently checked the literal gate lists, not every incoming semantic theorem. The broader frontier also retains the inherited fifteen-positive 66-operation/degree-157 equation, 75/44, and source 56's 73/54 and 74/46. The new short variants do not dominate all of those points. The two twelve-positive points above remain relevant and are not surpassed by merely comparing with the older degree-897 equation.

Ideas combined, not merely recopied. Degree-separated kernels allow linear and quadratic labels to collide safely because they multiply different powers of the packed word. Positive coordinate gaps delete additional coefficient records. A bit whose value is already forced by a recovered congruence can be removed from a carry mask, simultaneously repairing the residue of a leading-block-free code. Finally, coprimality and a fixed congruence distinguish two square classes without a large separator, replacing an eight-factor norm by a quartic norm. The nine-positive branch instead combines the free carrier with the previous sharper quotient majorant and a global addition-chain plan for all base powers.

Formal status. Commit `dabf4e39f1039a7f77056517ecbb01d6db1a8ced` advances the formal proof of source 52: its literal source gates, residuals, shifted records, selective bounds, low-mask decoding, cofinal bounds, ratio substitution and margin-first finalizer have formal interfaces. The pinned status still distinguishes these pieces from an assembled new source adapter with actual native-port calibration and an internal object-ZFC equivalence. This contribution does not claim

that those formal results automatically certify its changed source, carry code or final circuit. Five directly pinned status/proof documents are retained for the precise boundary.

3 Gap coordinates reduce the quadratic source

The incoming degree-separated source has 31 positive coordinates and 22 quadratic residuals, with 89 coefficient records after its two invertible row operations. A positive-coordinate change reduces this to 86 records without changing the finite positional certificate.

Write $q = 1 + \beta J$, $k = \eta + \zeta$ and $Y = a - E$. The source equations give

$$Y = sN > 0, \qquad E = XY > 0, \qquad c - \eta = kY > 0, \qquad P - k = E(c - \eta) > 0.$$

Therefore replacing old P by $T + k$ is a bijection on positive zero sets. The affected residuals are

$$T - Ec + E\eta, \qquad \tau^2 - T^2 - \eta T - \zeta T - 1.$$

The first loses two records and the second keeps its support. A further one-record reduction is obtained either by replacing old N by $n + 1$, or by replacing old L by $\ell - 1$. These are alternatives, not simultaneous changes. Their inverse maps preserve positivity because $N = q^3 \geq 8$ and $\ell = q^2 \geq 4$, respectively.

The same 116-maximum-weight certificate passes all interactions: 22 481 for the N-gap variant and 22 451 for the L-successor variant. The kernel degrees become $(1719, 2020, 1929)$, lowering the linear degree from 2027; convolution height remains 2161. The quadratic leading coefficient stays $+1$. Thus the inherited unconditional positivity estimate is strengthened, not weakened. Explicit source polynomials, kernels, and an independent symbolic reconstruction are included.

An optional source lemma reduces the positivity guards from twelve to eight: $F, \alpha, z, h, s, w, \eta, \zeta$. With native $M_C, M_F \geq 0$, transport first forces $C = q - F - \alpha > 0$, and

$$R = (q^2 - 1)((q - 1)C + q\alpha) + (M_C + qM_F)J > 0.$$

Hence $j \geq 1$ from $R = jc - V$. The identity $k = jc - V + 1 + hE$ and $c = kY + \eta > k$ exclude $i = 0$ and $g = 0$: in either case the remaining equations force $V \leq 1$, which would imply $k \geq c$. If $\gamma = 0$, then $k > E$ gives $c > X$ and $D = X + ac < (a + 1)c$, contradicting $D^2 = ((a + 2)^2 - 1)c^2 + 1$. The other coordinate positivities follow directly from their defining equations. This lemma is not advertised as a gate-count saving: the degree-separated encoder already reuses its full positive word in the low digit mask.

A further alternative replaces old N by $n + q$ simultaneously with the P-gap change. Its inverse is $n = N - q = q(q^2 - 1) > 0$, so this is again a positive-zero-set bijection and leaves only 85 coefficient records. Unlike the 86-record alternatives, this source needs a new positional layout: the former layout has exactly one unintended target interaction. Only a separately audited replacement certificate may be used with this 85-record version.

4 Repacked homogeneous layers and positive leading coefficients

The source coordinate changes admit three frozen positional certificates. The following table records source-code heights, not final equation degrees.

Certificate	Records	W	Q	(d_0, d_1, d_2)	H
Gap N	86	116	2036	$(1571, 1927, 1836)$	2068
Gap $N - q$	85	116	2050	$(1627, 1941, 1850)$	2082
Reflected gap N	86	116	2031	$(1935, 1941, 1825)$	2057

Here $d_r = \deg D_r$, $Q = \max q_j$, and $H = \max_{0\le r\le 2}(d_r + rW)$. The incoming split116 height was 2161. These are certificate-backed improvements, not claims of optimal packing.

For a source monomial x^α of degree r, let $e(\alpha) = \sum_i \alpha_i w_i$. A coefficient $a_{j,\alpha}$ in row j receives weight $2a_{j,\alpha}$ for degrees zero and one or a diagonal square, and $a_{j,\alpha}$ for a quadratic cross term. The reversed kernels and positive-coordinate word are

$$D_r(T) = \sum_{j,|\alpha|=r} c_{j,\alpha} T^{q_j - e(\alpha)}, \qquad U(T,x) = \sum_i x_i T^{w_i}.$$

The independent verifier reconstructs all 22 target polynomials in all 31 witnesses and all four independent coefficient ports. Its finite isolation check establishes

$$[T^{q_j}]\big(D_2(T)U(T,x)^2 + D_1(T)U(T,x) + D_0(T)\big) = 2\rho_j(x).$$

There are 22,481 interactions for either 86-record source and 22,450 for the 85-record source. The latter needs the new targets: its former incoming layout had an extra cross-row collision and is not reused here.

A simpler positivity argument. The reflected certificate replaces every old position w_i by $117 - w_i$ and negates source row 2. All positions remain distinct and positive; a row sign change preserves the common zero set. Its leading coefficients are

$$\mathrm{lc}(D_0) = 2K_0, \qquad \mathrm{lc}(D_1) = 2, \qquad \mathrm{lc}(D_2) = \beta.$$

They are all positive integers. If M strictly bounds every absolute kernel coefficient and $B > M + 1$, a polynomial of degree d with positive integral leading coefficient is greater than

$$B^d - M\frac{B^d - 1}{B - 1} > 0.$$

Thus each $D_r(B) > 0$ separately and $G(B,U) = D_2(B)U^2 + D_1(B)U + D_0(B) > 0$ for every $U \ge 0$. Unlike the earlier proof, this does not require the quadratic layer to have strictly greater numerical height than the linear layer.

There is a consequential leading-term tie:

$$d_2 + 2W = d_1 + W = 2057.$$

For $U = \sum_i B^{w_i} + g$, the leading B-coefficient of G is $\beta + 2$, *not* β. The exact final degree certificates account for this positive sum.

Bounds and completeness. The coherent original31 tuple admits the conservative bound $M = 2^{2134927}$ and $B = 4096Mb^2$. The coefficient audit must be repeated for whichever complete tuple is actually bound; ports are not mixed. Writing $\ell = \sum_i B^{w_i}$, distinct positions give $\ell < 2B^W$. On $0 \le g < 4bB^W$, put $U = \ell + g$ and $Z = B^{H+1}$. A geometric-series coefficient bound gives

$$0 < G(B,U) < \frac{43Mb^2B^{H+1}}{B-1}, \qquad \frac{2G(B,U)}{Z} < \frac{1}{32}.$$

The reflected target gap $H - Q = 26$ gives, for $A = \sum_j B^{q_j}$, $A < Z/64$ and $(B-2)A < Z/64$. Therefore with

$$N_0 = B^{W+1}, \quad M_0 = N_0 - 1 - (b-1)\ell, \quad J_0 = 2G + A,$$
$$S = g + N_0 J_0, \qquad T = M_0 + N_0(B-2)A,$$

one has $S + T < N_0 Z$. These inequalities concern arbitrary bounded g, before digit recovery. Once the low mask yields $g = \sum_i n_i B^{w_i}$, $0 \le n_i < b$, the word U encodes $x_i = n_i + 1$. All convolution coefficients have absolute value at most $993Mb^2 < B/4$, so the inherited sparse $+1$ signed-borrow test applies.

Every sufficiently large admissible base represents a fixed positive source zero. Row 5 still has value 1 at the all-one tuple, so some digit n_i is nonzero. Since all positions are positive, its encoded word is at least B and is divisible by B. These facts preserve the lower-bound and divisibility conditions used in the integer composition.

The unreflected certificates instead have quadratic dominance gaps $(25, 497)$ and$(25, 455)$ and positive quadratic leading coefficient 1; the usual geometric-series comparison proves their positivity. Read-only checks, byte-identical kernel rebuilds and coherent-hash tamper controls accompany all three certificates. They authenticate finite data and polynomial identities, not the inherited unbounded compiler theorem or an end-to-end kernel proof.

5 Nine positive unknowns: refined free-carrier composition

The source transformations and positional certificates above give three layouts of heights $2068, 2082, 2057$, each with $W = 116$. All fixed coefficients are the complete original31 tuple; the high mask is already shifted. The two finalizers below give the following operation/exact-degree pairs.

Layout	source records	simple finalizer	refined finalizer
$H = 2082$	85	444/1 845 940	476/1 520 178
$H = 2068$	86	449/1 834 516	481/1 510 770
$H = 2057$	86	451/1 825 540	483/1 503 378

These six points are nondominated within the delivered family. The cheapest and lowest-degree recommendations improve the incoming nine-positive point 473/1 910 404 in different ways. There is no global minimality claim. The natural-domain translations cost seven further operations each and have the same exact degrees.

5.1 The literal calibrated source

The three kernels D_0, D_1, D_2, labels w_i, and targets q_j are the literal data in the selected `nine/data/H*/` directory. They satisfy the proved coefficient-extraction identities for the corresponding positive source. Put $M = 2^{2134927}$, $\kappa = 2134939$, and set

$$B = 2^\kappa b^2, \qquad \ell = \sum_i B^{w_i}, \qquad A_t = \sum_j B^{q_j},$$

$$U_s = \ell + g, \qquad G_s = D_2(B)U_s^2 + D_1(B)U_s + D_0(B),$$

$$N_0 = B^{W+1}, \qquad E_c = H + W + 2, \qquad M_s = N_0 - 1 - (b-1)\ell,$$

$$S_c = g + N_0(2G_s + A_t), \qquad T_c = M_s + N_0(B-2)A_t.$$

Here $H = \max_d(\deg D_d + dW)$ is a fixed height, not a variable. In the unreflected layouts the positive leading quadratic kernel dominates the two others. In the reflected layout all three kernels are independently positive for $B > M + 1$. Its quadratic and linear heights tie, and their leading coefficients add to $\beta + 2 > 0$. Thus $G_s > 0$ before a final zero or a binary-base hypothesis is assumed.

On the branch $g < 4bB^W$, the exact geometric estimates give $2G_s/B^{H+1} < 1/32$, while each target contribution is below $1/64$ of that block. Hence $S_c + T_c < B^{E_c}$. The sparse borrow test yields

$$S_c \,\&\, T_c = 0 \quad \Longleftrightarrow \quad g \text{ encodes a positive common source zero}$$

when b is a power of two. This uses the coefficients bounded by $993Mb^2 < B/4$ and the target digit $4\rho_j + 1$ tested against $B - 2$.

Let $H_p = E_c + W + 23$, and define the fixed integers

$$\delta = (H_p - 15) \bmod 2, \qquad \lambda = (22 - H_p\kappa) \bmod 2,$$
$$e_N = (H_p - 15 - \delta)/2, \qquad e_2 = ((15 + \delta)\kappa + \lambda - 22)/2.$$

Use the polynomial registers

$$L = 2^{\lambda} b B^{E_c}, \qquad N = 2^{e_2} b^{\delta} B^{e_N}, \qquad R = L(S_c + T_c + 1) - S_c - 1, \qquad t = bB^W.$$

The exact two-block population identity proves

$$N^2 \mid \binom{2R}{R} \quad \Longleftrightarrow \quad S_c \,\&\, T_c = 0.$$

Indeed, if $b = 2^k$ and $B = 2^v$, then $\operatorname{pop}(T_c) = (W + 23)v - 31k - 22$ and $\log_2 L + \operatorname{pop}(T_c) = 2\log_2 N$. Unconditionally $R, N \geq 8$ and $b \leq R, N$. The endpoint $b = 1$ is harmless: its low mask would force $g = 0$, contrary to the positive domain.

5.2 The nine-input equation

The nine inputs are exactly $b, g, h, s, w, C, i, j, n > 0$. Set $j_0 = j - 1$ and

$$\begin{aligned}
U &= N^2 w, & Y &= N^2 s, & M_0 &= RY, & P &= 2M_0^2 U,\\
K &= R + 1 + h(P - 1), & A &= M_0(U + 1), & B_1 &= 2R + 1, & W_1 &= bw,\\
J &= 4A - 5, & Q &= 3W_1 C - 2(W_1^2 - 1), & D &= (A^2 - 1)C^2 + 1,\\
k_0 &= iJC, & E &= k_0 DC, & F &= (A^2 - 1)E^2 + 1,\\
\Lambda &= CD + 1 - A, & G &= A + F\Lambda, & H_1 &= C + F(B_1 + j_0 C),\\
I &= (G^2 - 1)H_1^2 + 1, & a_0 &= (P^2 - 1)K^2 + 1, & a_1 &= DFI,\\
m &= K^2(4t - g) - 16t(C - KY)^2, & \Omega &= Jm, & V_0 &= PK,\\
E_0 &= J(n + W_1^2) + Q.
\end{aligned}$$

All symbols are polynomial abbreviations, not extra quantified unknowns. For the refined version put

$$\begin{aligned}
B_8 &= 8D^2 - 4D - 1, \qquad N_* = 2G + \Lambda,\\
T_6 &= k_0\{128B_8N_* - k_0^2[16(B_8 + D) - 8]\Lambda + C(14k_0^2 - 192)\},\\
T &= T_6 + i^3\{C[C(576A - 2151) - (896A^2 - 4256A + 8442)] + 1024\} + 288i + 2048,\\
V_1 &= TH_1, \qquad q_* = 2048^2, \qquad \Theta = E_0 - \Omega(V_0 + V_1).
\end{aligned}$$

The single equation is

$$\boxed{\left[\Theta^2 - \Omega^2(a_0 + q_* a_1)\right]^2 - 4q_*\Omega^4 a_0 a_1 = 0.} \tag{1}$$

The cheaper simple version replaces V_1 by $k_0(8D^2 - 4D - 1)GH_1$ and q_* by 64.

5.3 Why a free carrier preserves the equivalence

The refined majorant proof retained from the prior construction needs only $A \geq 729$, $C \geq 1$, and $i \geq 1$. Here $A \geq 33280$, so it applies without the old carrier substitution and gives

$$V_0 > \sqrt{a_0}, \qquad V_1 > 2048\sqrt{DFI} > 0.$$

Also

$$E_0 = Jn + (J-2)W_1^2 + 3W_1C + 2 > 0.$$

A zero of (1) cannot have $m = 0$. Its real two-root factorization gives

$$E_0/J = m(V_0 + V_1 + r_0 + r_1), \qquad r_0^2 = a_0, \quad r_1^2 = q_* a_1.$$

The strict majorants make the parenthesis positive for every sign choice; therefore $m > 0$ is obtained *before* any integrality assertion. It follows that $|C - KY| < K/2$, hence $C > 2K > 2R + 1 + W_1$. This recovers the old carrier range. The usual ratio inequalities now give $a_1 > a_0$, so $r_0 + r_1 = \Theta/\Omega$ is rational and nonzero. Squaring isolates both roots, which are rational and consequently integral; $r_1/2048$ is integral as well. The displayed factor identity gives $J \mid Q$. The exact CRT replacement using $F \equiv 1 \pmod{JC}$ therefore supplies all four inherited ratio conditions: the two squares, divisibility, and positive margin. Jones's theorem and the source decoder yield a syntactic refutation.

Conversely, take the ratio witnesses supplied by a source zero and the positive roots. Choose

$$n = m(V_0 + V_1 + r_0 + r_1) - W_1^2 - Q/J.$$

It is integral. Since $A > W_1$, $D \geq A^2$, $D > C^2$, and $V_1 > 2048D$, one has $W_1^2 + Q/J < 2D < V_1$, so $n > 0$. One norm factor vanishes. This proves the existence equivalence and, by negation, the ZFC-provable equivalence between no positive zero and Con(ZFC). The inherited compiler and numerical ratio theorem remain identified dependencies; no new end-to-end kernel certificate is claimed.

5.4 Exact degree and implementation checks

Let $r = \deg R$, $\nu = \deg N$, $c_0 = 2W + 1$, and $a = r + 4\nu + 2$. The exact local defect $T^2 - 2048^2 DFG^2$ has703 terms, each with exponents at most $(15, 18, 6)$ in (A, C, i). The unique term attaining all three maxima is $9 \cdot 2^{18} i^6 A^{15} C^{18}$. Its complete integer expansion is independently reconstructed. Consequently, for $\Delta_1 = V_1^2 - q_* a_1$,

$$\Delta_1 = H_1^2(T^2 - q_* DFG^2) + q_* DF(H_1^2 - 1), \qquad \deg \Delta_1 = 31a + 2r + 44.$$

Resolve the main cancellation by the exact identity

$$\begin{aligned}\Theta^2 - \Omega^2(a_0 + q_* a_1) = E_0^2 - 2E_0\Omega(V_0 + V_1) \\ + \Omega^2[(K^2 - 1) + \Delta_1 + 2V_0V_1].\end{aligned}$$

The term $\Omega^2\Delta_1$ uniquely dominates, and its square dominates the remaining norm term. Thus the refined exact degree is

$$\boxed{86r + 328\nu + 4c_0 + 260}, \qquad r = 4E_c - 1, \quad \nu = 2e_N + \delta.$$

Its diagonal leading coefficient is $81 \cdot 2^{88} \tau^4 \rho^{86} \sigma^{328} > 0$, where ρ, σ, τ are the leading coefficients of R, N, t. The simpler finalizer has degree $104r + 400\nu + 8W + 316$.

The evaluator charges every binary addition, subtraction, and multiplication, including multiplication by constants. The new global backward addition chains share required base powers across all three kernels, the masks and the threshold. They do not treat exponentiation as one operation. The package includes all12 literal circuits, their fixed coefficient payloads, 1080 independent actual-coefficient modular replays, full coefficient-recipe checks, exact degree inequalities, and the exact703-term identity.

6 A congruence-separated quartic norm in ten integers

The ten-integer route combines the new positive-gap, homogeneous-degree source with the two-root congruence finalizer of the incoming *Congruence Norms* contribution. A further one-bit deletion removes the outer leading block without losing the congruence required by the arithmetic bridge. The result is an ordinary integer polynomial, not an exponential Diophantine formula. All named registers below are abbreviations, not additional unknowns. The exact coefficient payloads and literal circuit are included in `integer/`.

6.1 The leadless source and its calibration

Use the gap source and its exact three-kernel certificate from the source section. Write $W = \max_i w_i$, $H = \max_d(\deg D_d + dW)$, and $A_T = \sum_j B^{q_j}$. Supply exactly

$$t, g_0, h_0, k_0, l, w, x, y, z, m \in \mathbb{Z}.$$

In the primary mirror layout, row2 is additionally negated, an invertible row operation; the kernel identity uses these oriented source rows. Set $\kappa = 2134940$, $\lambda = 2^\kappa$, and define

$$\begin{aligned} b &= 6t+4, \quad g = bg_0, \quad h = bh_0, \quad k = b(3k_0+2), \quad B = \lambda b^2, \\ \ell &= \sum_i B^{w_i}, \quad u = \ell + g, \quad N_0 = B^{W+1}, \quad N = B^{W+H+2}, \\ G_c &= D_2(B)u^2 + D_1(B)u + D_0(B), \quad M_c = N_0 - 1 - (b-1)\ell, \\ S_c &= g + N_0(2G_c + A_T), \quad T_c = M_c + N_0(B-2)A_T - 2. \end{aligned}$$

The source ports for the bridge are

$$\boxed{X = N(S_c + T_c + 1) - S_c - 1, \qquad Y = 2^{16\kappa-23} b B^{H+2W+9}, \qquad \mathcal{C} = bB^W.} \tag{2}$$

No division appears in these expressions. The coefficient $2^{16\kappa-23}$ is materialized, not left as an unevaluated algorithmic parameter.

All signed assignments have $b \neq 0$, $2 \mid b$, and $b \equiv 1 \pmod 3$. Because all source weights are positive, direct reduction proves

$$X \text{ odd}, \qquad b \mid X+1, \qquad b \mid Y, \qquad 3 \mid X.$$

Indeed $B, N, N_0 \equiv 1 \pmod 3$, $\ell \equiv 31$, $A_T \equiv 22$, so the old high mask is $-22 \equiv 2$, the new T_c is zero, and $X \equiv T_c$.

The deletion of 2 from the mask is justified only after the finalizer has recovered that b is a positive even square. At that point $b \geq 4$ and $4 \mid g$. The low base-B digit of the old mask is $B-1$, while bit1 of S_c is zero. Thus clearing bit1 preserves binary disjointness and reduces the mask population by exactly one. The usual source bounds give $S_c + T_c < N$ for $0 < g < 2\mathcal{C}$.

If $b = 2^s$ and $B = 2^v$, then

$$\operatorname{pop}(T_c) = (W+23)v - 31s - 23.$$

The two disjoint blocks of $X = (S_c + T_c)N + (N - 1 - S_c)$ give

$$\operatorname{pop}(X) = \log_2 N + \operatorname{pop}(T_c) - \big(\operatorname{pop}(S_c) + \operatorname{pop}(T_c) - \operatorname{pop}(S_c + T_c)\big).$$

The maximum is $(H + 2W + 25)v - 31s - 23 = \log_2 Y$. Consequently

$$Y \mid \binom{2X}{X} \quad \Longleftrightarrow \quad S_c \,\&\, T_c = 0,$$

and the source certificate decodes a positive common source zero. This is an exactly recalibrated threshold, smaller by four than the old leading-block threshold. It does not assume $4 \mid b$ on arbitrary signed inputs.

6.2 The complete polynomial

Define the division-free bridge registers

$$\begin{aligned}
L &= lY, & U &= 2LX, & V &= 4gwY, \\
W_b &= bw, A = U(V+1), & B_1 &= 2X+1, & C &= B_1 + (A-2)h, \\
P_{1/2} &= (LX)UV, & P &= 2P_{1/2}, & K &= X+1+k(P-2), \\
S &= 2A-5, & T &= 3W_bC - 2(W_b^2-1), & Q &= (P_{1/2}^2-1)K^2+1,
\end{aligned}$$

$$\Delta = 8\mathcal{C}^3 gK^2 - g^2\big(32(C-KL)^2\mathcal{C}^3 + g^2K^2\big).$$

The subscript $1/2$ names an explicitly multiplied polynomial; it is not an instruction to divide an unknown. Put

$$\begin{aligned}
\xi &= zx, & D &= (A^2-4)C^2+4, & E &= C^2D\xi, \\
F &= 4(A^2-4)E^2+1, & G &= 2(A+2)(A-2)^2E^2-1, & & \\
\Omega &= 16E, & H_c &= C + (B_1-C)F + \Omega Fy, & I &= (G^2-1)H_c^2+1, \\
\Sigma &= \xi S^5, & \rho &= T + Sy, & \Theta &= (\rho^3 - \rho S^2 - S^3)(\rho^2+23S^2), \\
r &= bQ, & a &= r + (3\Delta-1)(rz)^2, & d &= DFI, \\
Z &= \Sigma m - \Theta. & & & &
\end{aligned}$$

The final equation is

$$\boxed{F_{10} = (Z^2 - \Sigma^2(a+d))^2 - 4\Sigma^4 ad = 0.} \tag{3}$$

The four fixed compiler ports are inherited as a whole from the original31 instance. The positive-gap transformations change its coordinates, not its coefficient binding.

6.3 Why the two-root norm is sufficient

The normalizations imply $2b \mid U, A$, $b \mid K$, $3 \mid K$, and $Q \equiv 1 \pmod{b^2}$, $Q \equiv 1 \pmod 3$. Also $2b \mid C+1$ and $4b \mid D$. Therefore, without any positivity assumption,

$$a/b \text{ is odd}, \qquad d/b \text{ is even}, \qquad a \neq d.$$

The odd integer S is never zero. The quintic $\pi(v) = (v^3 - v - 1)(v^2+23)$ has no rational root, so $\Theta \neq 0$. At a zero of (3), $\Sigma \neq 0$. With $q = Z/\Sigma$, the elementary identity

$$(q^2 - a - d)^2 = 4ad$$

and $a \neq d$ force $q \neq 0$. The rational numbers

$$u_0 = \frac{q^2 + a - d}{2q}, \qquad v_0 = q - u_0$$

satisfy $u_0^2 = a$, $v_0^2 = d$. They are integers because their squares are integers. Hence both radicands are squares and $\Sigma \mid \Theta$; the standard valuation argument gives $S \mid T$.

The coprime factors of $a = r[1 + (3\Delta - 1)rz^2]$ are signed squares. Since $r \equiv 1 \pmod 3$, they must both be positive squares. The same argument for $r = bQ$, using $\gcd(b, Q) = 1$ and $b \equiv Q \equiv 1 \pmod 3$, makes b, Q positive squares. Then $b \geq 4$, and the second factor of a forces $\Delta > 0$. The definition of Δ gives $K \neq 0$ and $0 < g < 2\mathcal{C}$. If $l = 0$, then $P = 0$ and $Q = 1 - K^2 \leq 0$, a contradiction. Thus the nonvanishing condition required by the inherited Sun bridge is recovered rather than imposed as an additional polynomial factor. That bridge now gives the central-binomial divisibility and hence a source zero.

For completeness, choose any sufficiently large $b = 4^s$ in the source encoding. Positive source weights give $b^2 \mid g$ and $X + 1 \equiv -g \pmod{b^2}$. The constructive bridge supplies the exact Lucas values $C = u_{2X+1}(A)$ and $K = u_{X+1}(P)$. The recurrence and the facts X odd, $4b^2 \mid P, A^2$ show that its existing witnesses satisfy $b \mid h$, $b \mid k$, and $k/b \equiv 2 \pmod 3$. Thus the displayed restrictions lose no canonical solutions. Pell's theorem for the positive nonsquare $(3\Delta - 1)r$ chooses z making a square. The inherited residue interpolation chooses ξ divisible by z and a suitable y while keeping d square and making $\Sigma \mid \Theta$. Then $m = \Theta/\Sigma + \sqrt{a} + \sqrt{d}$ makes (3) vanish.

These arguments compose with the source/refutation theorem to establish, in ZFC, that the absence of integral solutions is equivalent to Con(ZFC). The complete soundness and completeness proof, including every source bound and congruence-class reconstruction, is in `integer/PROOF.md`. The bridge references are Sun's 2021 Theorems4.12–4.13 and the modified certificate/interpolation results in his 2026 ten-unknowns preprint. They remain explicit mathematical dependencies, not claims of newly checked end-to-end kernel verification.

6.4 Exact degree and evaluator accounting

Write $r_X = 4(W + H) + 6$, $r_Y = 2H + 4W + 19$, and $a_A = r_X + 2r_Y + 4$. The dominant final term is uniquely $\Sigma^4 d^2$, and

$$\deg d = 88a_A + 126, \qquad \deg \Sigma = 5a_A + 2.$$

Thus the exact total degree is

$$\boxed{\deg F_{10} = 196a_A + 260 = 2352W + 1568H + 9668.}$$

Full literal circuits, live-gate counts, actual-tuple modular replay, and nonzero highest-degree coefficients over three primes independently check the stated metrics. Global shared power chains reduce evaluation gates without changing a single polynomial coefficient. The fixed threshold is materialized as one leaf; charging the five constant-only gates in the older evaluator convention adds five operations, with no effect on degree.

7 Smaller evaluators for the integral-root equations

The incoming integral-root mechanism is retained, rather than advertised as a new discovery here. Three literal fifteen-positive circuits improve from operation/degree pairs 90/24, 88/26 and 79/30 to

$$(88, 24), \qquad (87, 26), \qquad (78, 30).$$

All retain the entire incoming 590069-bit-radix tuple, not the original31 tuple used by the nine- and ten-variable branches.

7.1 The explicit degree-24 polynomial

The positive inputs are

$$j, F, \alpha, z, f, h, i, T, X, Y, r, \eta, \zeta, u, \gamma.$$

Define the following polynomial abbreviations:

$$\begin{aligned}
J &= 3j+1, & q &= \beta J + 1, & C_w &= q - F - \alpha,\\
E &= XY, & k &= \eta + \zeta, & c &= kY + \eta,\\
a &= E + Y, & H &= 4a + 3, & \Delta &= a^2 + H,\\
D_0 &= X + ac + \gamma H, & \ell &= EkY, & I &= ic^2,\\
R &= ((q-1)(q-F)+\alpha)(q^2-1) + (M_C + qM_F)J,\\
\widehat{R} &= k - hE - 1, & U &= \Delta(f^2-1), & V &= f^2(cT - \widehat{R}) - c.
\end{aligned}$$

The residuals are

$$\begin{aligned}
A_0 &= [q^3(\ell + r) + Y]^2 - q^6[\ell(\ell + k) + 1],\\
A_1 &= (qD_0 + XH)^2 - q^2(\Delta c^2 + 1),\\
A_k &= \widehat{R} - R,\\
A_t &= q[K_0 C_w - F - z(q-1)] + XC_w,\\
A_a &= (V+u)^2 - Uu\big(V + (V+u)\big) - 1,\\
A_s &= I^2 - U.
\end{aligned}$$

Then the completed equation is

$$\boxed{A_0^2 + A_1^2 + A_k^2 + A_t^2 + A_a^2 + A_s^2 = 0.}$$

Its literal evaluator, with common subexpressions shared, has 88 live binary gates. The highest homogeneous part remains

$$4X^4Y^6f^8u^2\big((\eta+\zeta)T + hX\big)^2,$$

which is nonzero and has degree 24.

7.2 Why both changes preserve the theorem

First share the already computed auxiliary gap: $2V + u = V + (V + u)$. This is an identity over every commutative ring and saves one addition in each of the three circuits.

For the degree-24 case only, the old transport residual is

$$A_t^{\mathrm{old}} = q[K_0 C_w + (q - F) - z_{\mathrm{old}}(q-1)] + XC_w - q.$$

Our residual obeys the exact all-input identity $A_t(z) = A_t^{\mathrm{old}}(z+1)$, while no other residual uses this transport coordinate. Any positive new zero is therefore an old positive zero with $z_{\mathrm{old}} = z + 1$. The converse uses canonical witnesses, not a claim about arbitrary old zeros. The retained ordinary compiler proof gives

$$w = 2^R/q > q > F, \quad C_w > 0, \quad (K_0 + w)C_w - F = (z_{\mathrm{old}} - 1)(q-1).$$

The left side is positive, and $q > 1$, so the integral z_{old} is at least two. Thus $z = z_{\mathrm{old}} - 1$ is a valid positive new coordinate. All subsequent integral-root reconstructions leave that transport coordinate unchanged. This supplies completeness.

The old integral-root theorem recovers $q^3 \mid Y$ and $q \mid X$ by rational-root integrality in the first and main monic square equations. Its congruence $J = 3j + 1$ is retained: deleting it without an alternative proof of $\gcd(q, 3) = 1$ is not justified. The inherited canonical Pell gaps and signed rank argument then give the refutation equivalence. The full predecessor proof and complete coefficient binding are retained as inert sources. Translation by a constant preserves degree, and the other two modified outputs are polynomial-identical to their predecessors.

The authored checker validates all source hashes, literal gates and the two identities. It performs 360 exact signed and 360 modular whole-circuit regressions, plus 6000 local checks. These are implementation checks; the two universal identities and canonical positivity argument above are the proof of the changes.

8 A genuinely different source: one-stack proof certificates

The user explicitly permits abandoning both K215 and Jones's universal expression. The following construction does so: it recognizes a proof nondeterministically, rather than arithmetizing a deterministic enumerator. Its finite accepting paths are mathematically equivalent to a ZFC refutation. The local polynomial is fully materialized. A closed global compiler-input proposal is supplied, but has not been elaborated, kernel compiled, or expanded to count its final variables. In particular, the local equation alone is *not* a consistency equation.

8.1 Numerical calculus and stack

Use the numerical calculus C_{222} of the pinned repository, not its machine implementation. Its independent semantic calibration is

$$C_{222}(1) \iff \text{there is a ZFC refutation.}$$

The precise source lemmas are `KEndpoint.c222_iff_zf_refutation` and `ConstructibleChoice.zf_refutation_iff_zfc_refutation`. Put $I(a,b) = 2(a+b)(a+b+1) + 4b$, $A(a,b) = I(a,b) + 2$, $M(a,b) = I(a,b) + 1$, $E(a,b) = I(a,b) + 3$, $N(a) = I(a,1)$, and $X(a,b) = N(A(a, N(b)))$. The eight unconditional axiom codes are

$$\begin{aligned}
r_1 &= X(a, E(a,b)),\\
r_2 &= I(A(a, I(b,c)), I(A(a,b), A(a,c))),\\
r_3 &= I(E(a,b), I(M(c,a), M(c,b))),\\
r_4 &= I(A(a, A(b,c)), A(b, A(a,c))),\\
r_5 &= I(E(c,a), I(M(c,b), M(a,b))),\\
r_6 &= I(X(a, A(a,b)), b),\\
r_7 &= I(E(c,a), I(E(c,b), A(M(a,b), E(a,b)))),\\
r_8 &= I(I(I(c,u), a), I(I(a,c), I(b,c))).
\end{aligned}$$

The remaining guarded code $B(a,b,c)$ is defined by

$$\begin{aligned}
R(b,k,m) &= A(b, I(m, N(A(b, I(m, N(M(b,k))))))),\\
C(a,b,k,m) &= A(b, I(M(b,k), N(A(a, I(m,\\
&\qquad N(A(a, I(A(k, I(M(k,a), M(k,b))), R(b,k,m)))))))))),\\
B(a,b,c) &= N(I(I(M(a,b), A(c, M(a,b))),\\
&\qquad A(c, I(C(a,b,c,c), N(R(b,c,c)))))).
\end{aligned}$$

It is available when $c \neq a, b$. All formulas are also included as literal binary arithmetic DAGs in `stack_step.json`.

Encode a list by $\mathrm{Cons}(h,t) = 1 + (h+t)(h+t+1)/2 + t$, with empty list 0. Every positive natural number has a unique head and tail, and the tail is strictly smaller. Hence every natural number is a finite stack code. Define the polynomial

$$L(s,h,t) = 2s - 2 - (h+t)(h+t+1) - 2t.$$

Over natural numbers, $L(s,h,t) = 0$ is exactly $s = \mathrm{Cons}(h,t)$. The accepted stack is $\mathrm{Cons}(1,0) = 2$.

8.2 Three branches, five shared witnesses

With natural witnesses a, b, c, u, v, put

$$\begin{aligned}
G &= (c-a)^2(c-b)^2 - (u+1)^2, \\
H &= v^2 \prod_{j=1}^{8} (v - r_j)^2 \big((v - B(a,b,c))^2 + G^2\big), \\
T_{\mathrm{ax}} &= L(s', v, s)^2 + H, \\
T_{\mathrm{mp}} &= L(s, I(a,b), v)^2 + L(v, a, c)^2 + L(s', b, c)^2, \\
T_{\mathrm{gen}} &= L(s, b, c)^2 + L(s', A(a,b), c)^2, \\
T &= T_{\mathrm{ax}} T_{\mathrm{mp}} T_{\mathrm{gen}}.
\end{aligned}$$

All three factors are nonnegative. Thus $T = 0$ means exactly one of the following permitted transitions (possibly more than one): push truth or an axiom; replace adjacent top entries $I(a,b), a$ by b; replace top entry b by $A(a,b)$. For the guarded axiom, $G = 0$ implies $c \neq a, b$ because $u + 1 > 0$. Conversely, for $c \neq a, b$ choose $u = |(c-a)(c-b)| - 1$. The same u can serve as the fourth parameter of r_8 in a different branch: an existential disjunction does not require distinct witness slots.

Exact correctness of finite paths. Every stack reached from 0 contains only C_{222} theorems, by induction on the path length and the three inference rules. Consequently reaching 2 implies $C_{222}(1)$. Conversely, prove by induction on a derivation that its conclusion h can be pushed onto *any* pre-existing stack S, without changing its tail. An axiom is pushed directly. For modus ponens first execute the derivation of a, then that of $I(a,b)$, then apply the pop-two/push-one branch. For generalization execute the premise, then apply the third branch. Thus a derivation of 1 yields a path from 0 to 2. Duplicating a premise merely increases time; no history register, cursor, or saved proof list is required. This proof is finitary and formalizable in ZFC, so the inherited semantic calibration gives

$$\exists \text{ a finite } T\text{-path from 0 to 2} \quad \longleftrightarrow \quad \neg\,\mathrm{Con}(\mathrm{ZFC}).$$

8.3 Materialized local quartic and closed global proposal

The shared binary DAG has 382 operation nodes, seven inputs, and numeral leaves at most four. Its syntactic degree upper bound is 1,048,802. Inline affine arithmetic and introduce a nonnegative wire only at each nonlinear product. Every such product is a square or a product of nonnegative expressions; this is why the guard uses separate squares. There are 84 wire variables. Together with a, b, c, u, v, the local source has 89 natural witnesses. Its 85 quadratic residuals have a sum of squares of exact degree four with 3,108 monomials. There are 91 total natural variables: two exposed

stack states and 89 existential witnesses. These are *local* counts, not counts for the final consistency equation.

For the global closure take only three outside variables t, x, y, and $\beta(x, y, i) = x \bmod (1 + (i+1)y)$. Require

$$\beta(x, y, 0) = 0, \quad \beta(x, y, t) = 2, \quad \forall i < t \; \exists \mathbf{w} \in \mathbb{N}^{89} \quad Q(\beta(x, y, i), \beta(x, y, i+1), \mathbf{w}) = 0.$$

The CRT coding lemma supplies x, y for every finite stack sequence. Conversely the displayed conditions decode a valid finite path. No extra $t > 0$ condition is needed: $t = 0$ contradicts the two endpoints. The displayed bounded formula is not itself an ordinary Diophantine polynomial. The artifact `OneStackEquation.lean` specifies its closed constructor translation: remainder graph representations, finite composition, `Rep.boundedForall`, existential closure of (t, x, y), and finally `QuarticCompiler.compile`. This bypasses Jones entirely. Its primitive constructors are explicit data-valued representations in the pinned repository, not an unspecified appeal to MRDP.

The local result and path equivalence above are mathematical proofs. The new Lean constructor proposal remains uncompiled, and its complete output is unmaterialized and uncounted. It is therefore a promising alternate architecture, *not a claimed smaller completed consistency equation.*

9 Reproduction, provenance and limitations

The ZIP contains the editable paper, compiled PDF, exact arithmetic circuits, explicit signed coefficient payloads, source and packing certificates, retained proofs and licenses, and newly authored verification programs. Search executables and unrelated incoming programs are not required to evaluate or check a delivered equation. The incoming archives were inspected as inert data; no upstream Python module was imported or executed.

9.1 Three different levels of evidence

1. *Byte provenance.* The original six archives have exact Git blob and SHA-256 identities. Retained members are matched to those archives, and directly retrieved formal sources are checked by their own Git blob identity. The sealed release manifest checks every included file.

2. *Finite arithmetic certificates.* These reconstruct source coordinate substitutions, check every relevant degree-specific target interaction, rebuild literal circuits, verify complete fixed-port bindings, and use exact identities and actual-coefficient leading-term witnesses. Modular samples are regression tests, not the sole basis for a polynomial identity or an upper degree bound.

3. *Universal mathematics.* Positivity recovery, carry normalization, rational square-root integrality, square-class separation, CRT and canonical Pell reconstruction prove the new existence equivalences. Their identified compiler/calculus and number-theoretic dependencies remain inherited. No finite search establishes consistency.

The finite checks cover all 22 target polynomials over 31 witnesses and four independent coefficient ports, not just sample values of the huge constants. Source packing is checked at 22,481 interactions for the 86-record layouts and 22,450 for the 85-record layout. The integral-root simplifications include 720 full-circuit exact/modular regression comparisons and 6,000 local identity checks. The refined majorant has a separately reconstructed exact 703-term defect polynomial. The direct proof source is checked against its literal local polynomial and quartic expansion. Deliberately corrupted packing coefficients and target positions are rejected even after their code hashes are updated coherently; these tests are distinct from sealed-file hash rejection.

9.2 How to replay

From a clean extracted archive, run `python verify.py`. The driver first checks the sealed manifest and source provenance, makes an isolated copy, and runs only the enumerated authored checkers and builders. Its rebuild ledger compares generated artifacts byte-for-byte with the sealed outputs. Run `python negative_controls.py` for deliberate corruptions; a rejected byte tamper is recorded separately from a rejected mathematical certificate. Build the PDF with `bash report/build.sh`. PDF metadata can differ across TeX environments even when the mathematical text is unchanged.

The two small-variable branches intentionally use the complete original31 coefficient tuple; the short integral-root branch retains its different complete 590069 tuple. No individual port is borrowed from a smaller tuple. The coefficients are compressed losslessly for storage; the circuit definition uses their integer values. These storage choices are not a new coefficient-height record.

9.3 Claims deliberately not made

The report neither proves nor refutes Con(ZFC), establishes a global minimum, changes positive unknowns into unrestricted integer unknowns without an explicit transformation, nor presents the new full equations as kernel-checked. The direct proof-certificate local polynomial is not itself a consistency equation. The reported arithmetic-operation count excludes fixed numeral construction and says nothing about measured evaluation time with enormous integer values.

10 Further research questions

1. Instantiate and measure the direct proof-certificate trace compiler. A small local relation is not automatically a small final Diophantine equation; bounded-universal elimination and witness coding must be included in the count.
2. Optimize the arithmetic proof calculus itself, independently of a machine implementation. Junk formula codes are permissible if the numerical closure relation retains the syntactic refutation equivalence.
3. Search jointly over positive-coordinate changes, reversible row operations, degree-specific labels and target positions. Optimizing a positional code before simplifying the source can freeze avoidable records into the objective.
4. Develop a proof-producing multi-output addition-chain planner for the common radix powers. The current circuits are explicit and replayable, but the search is not a proof of global circuit optimality.
5. Determine whether the congruence-separated two-root finalizer can replace other high-degree square-class mergers while preserving unconditional signed soundness and canonical completeness.
6. Transfer additional divisibility conditions into existing monic roots without increasing their cleared-denominator degrees. The failed naive changes in the integral-root branch show why denominator cost and positive witness reconstruction must be analyzed together.
7. Formalize the changed source, carry and finalizer adapters against the existing Lean/Rocq endpoint. Exact finite certificates and modular degree witnesses do not replace the universal semantic proof.

Conclusion. The completed equations improve explicit tradeoffs. The separate direct proof source replaces enumeration by existential proof certificates, but its global trace elimination remains unmaterialized.